

CSC 360, Quiz 3 Practice

Consider the following code:

```
abstract class Person {  
  
    String first;  
    String last;  
  
    Person(String first, String last) {  
        this.first = first;  
        this.last = last;  
    }  
  
    public String returnA() {  
        return "AAA";  
    }  
  
    public abstract String returnB();  
  
}  
  
public class Student extends Person {  
  
    Student(String first, String last) {  
        super(first, last);  
    }  
  
    @Override  
    public String returnA() {  
        return "A";  
    }  
}
```

```

@Override
public String returnB() {
    return "B";
}

public static void main(String[] args) {
    Person p1 = new Student("Tommy", "Thompson");
    System.out.println(p1.returnA());
}
}

```

Question 1) What is the output of the above code?

Question 2) What would the output be if *returnB()* was called instead of *returnA()*?

Question 3) What would the output be if *returnA()* was called but Student did not override *returnA()*?

Question 4) What would the output be if *returnB()* was called but Student did not override *returnB()*?

Question 5) What would the output be if main was changed to the following?

```

Student p2 = new Person("Alice", "Smith");
System.out.println(p2.returnA());

```

Question 6) How would you modify the *Person* class so that a list of *Person* objects can be sorted alphabetically by last name? Hint: the *MyRange* class example creates objects that can be sorted.

Question 7) Based on the TicTacToe UML diagram (<https://tinyurl.com/423xksj9>), describe the relationship between the following classes. Your description should indicate whether inheritance, aggregation, or composition is used, and where an object of each class should be created.

- a) Player and CpuPlayer
- b) TicTacToe and Board